

Course

« *Computer Vision* »

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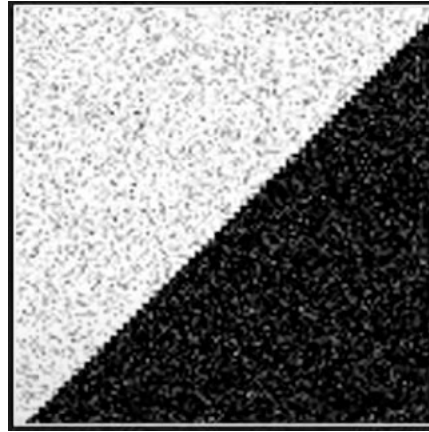


IX. Corner detection

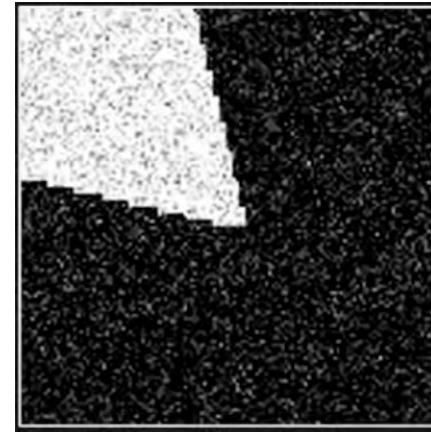
- What is a corner in an image?



A flat region



An edge region

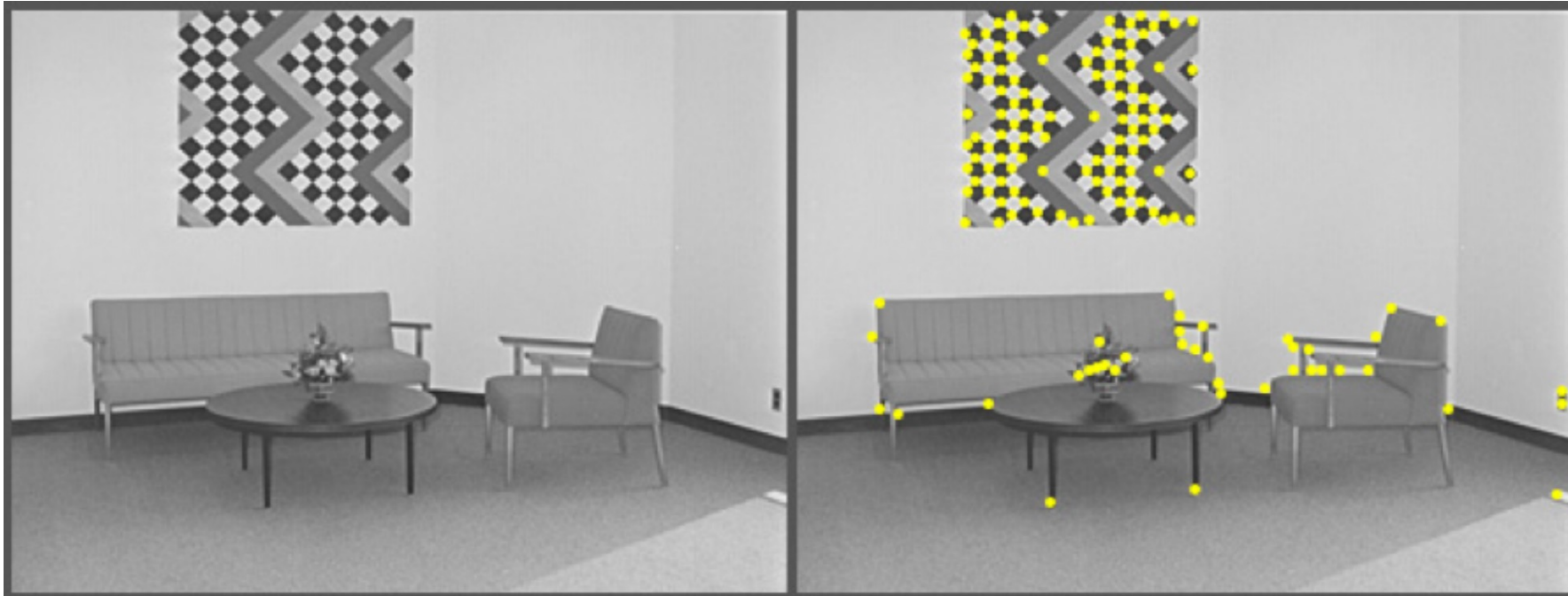


A corner region

- **A corner** is a point where two edges meet, that is where we have rapid changes of image intensity in two directions within a small region.

IX. Corner detection

- Edge detectors perform poorly at corners.
- Corners provide repeatable points for matching
=> So they are worth detecting!



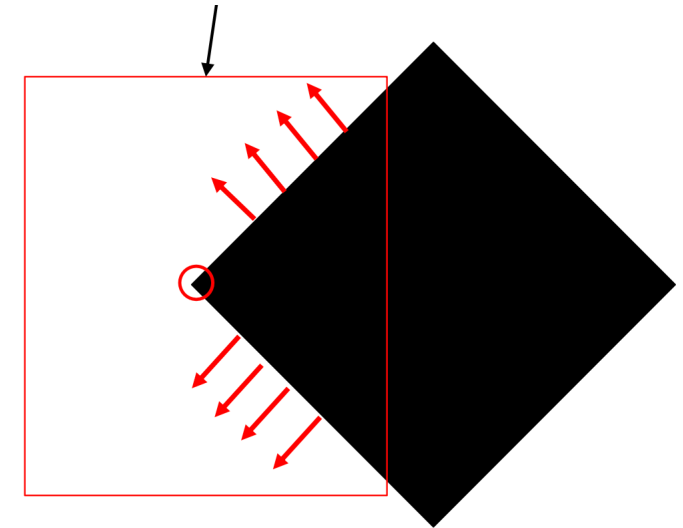
IX. Corner detection

How to detect a corner in an image?

General idea:

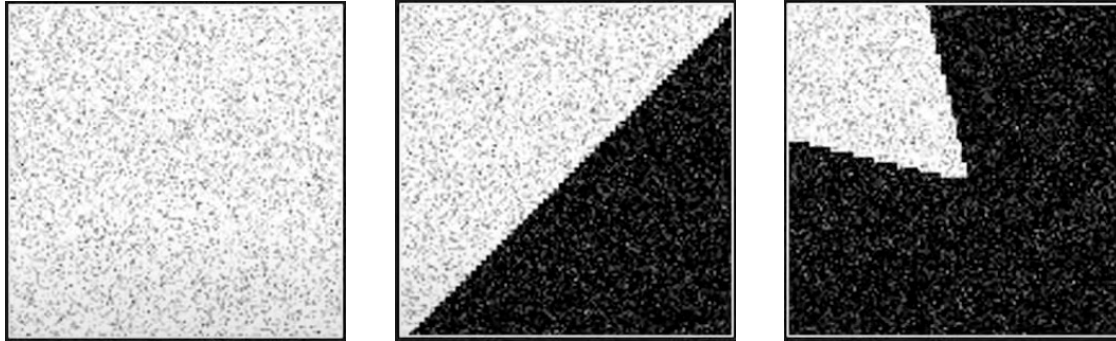
- Exactly at a corner, gradient is ill defined.
 - However, in the region around a corner, gradient has two or more different well-defined vectors.
- ⇒ As it is about changes in image intensity, corner detection also relies on derivatives.

Region around the corner

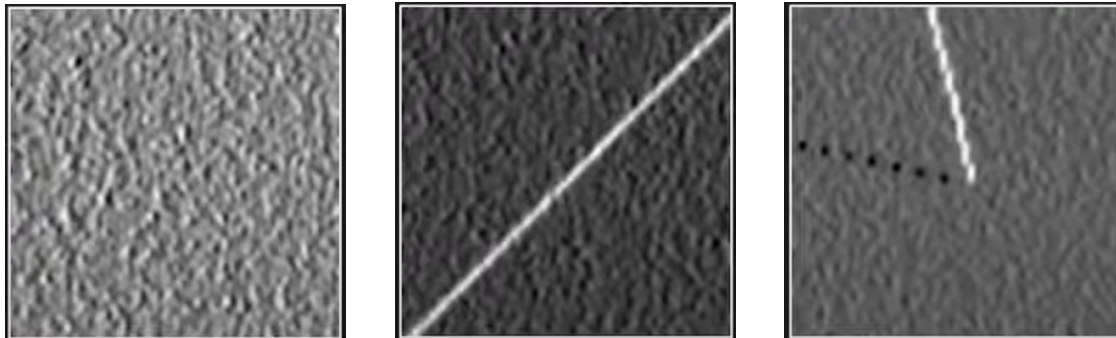


IX. Corner detection

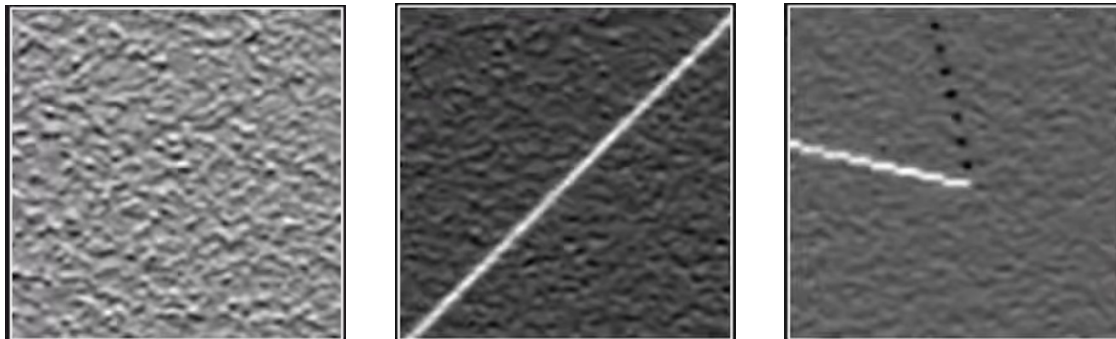
I



$$I_x = \frac{\partial I}{\partial x}$$

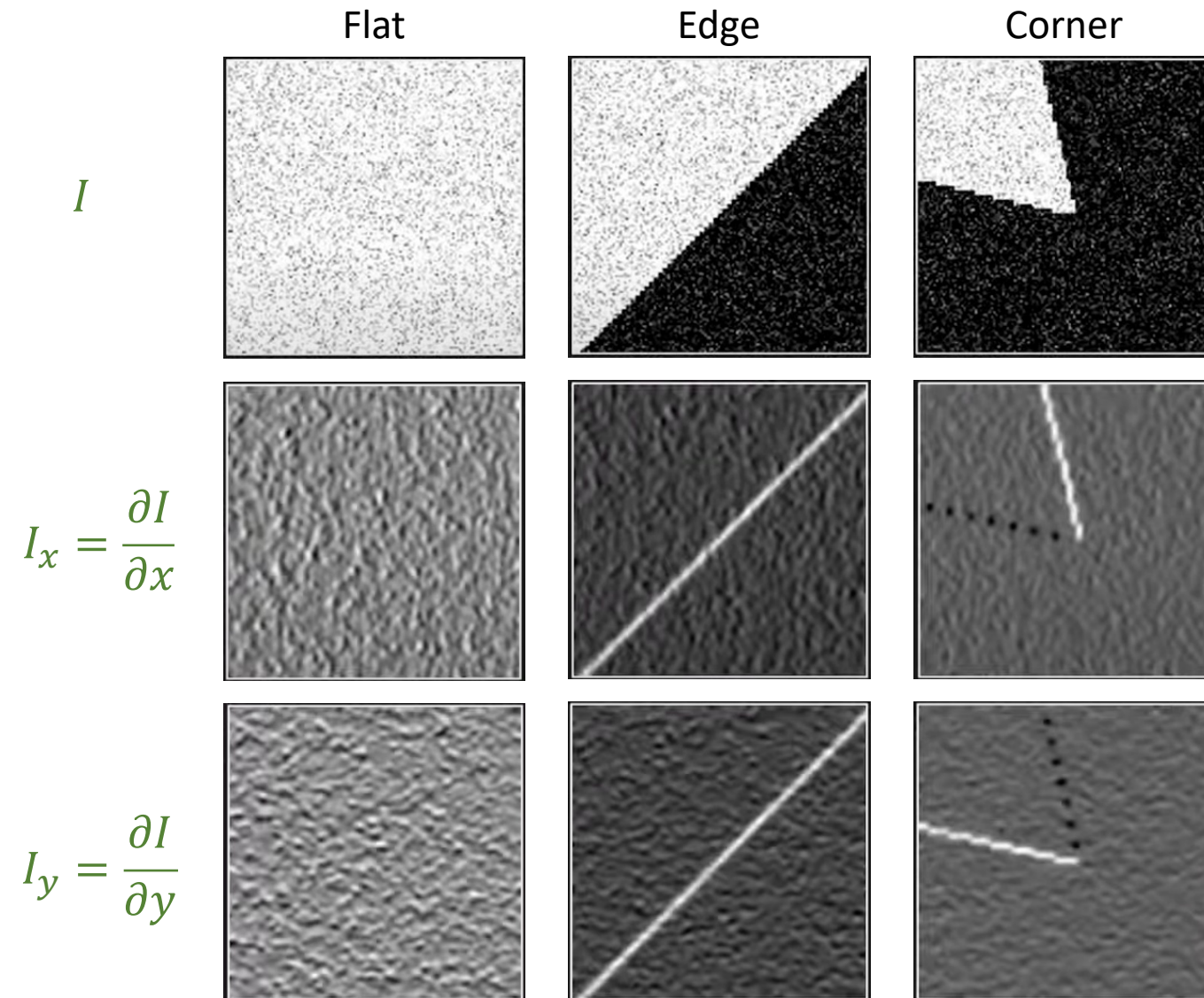


$$I_y = \frac{\partial I}{\partial y}$$



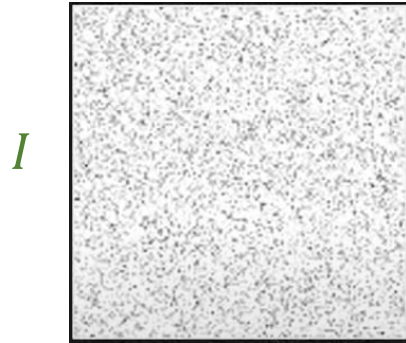
⇒ We classify these regions (flat, edge or corner) based on the distribution of the gradient.

IX. Corner detection

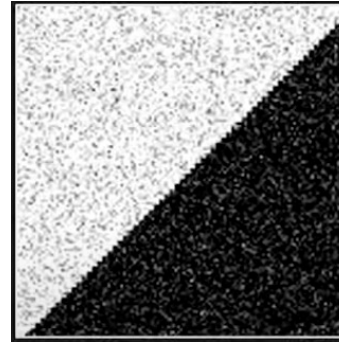


IX. Corner detection

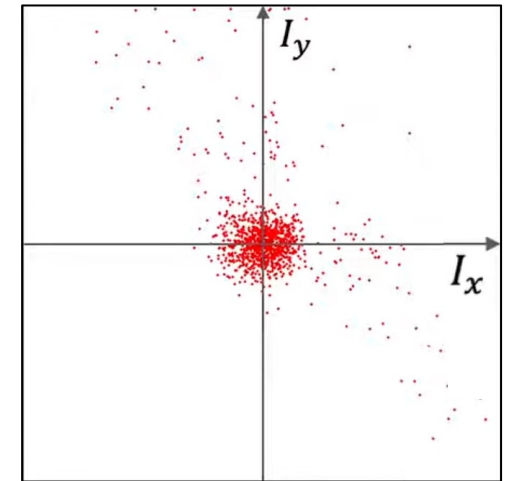
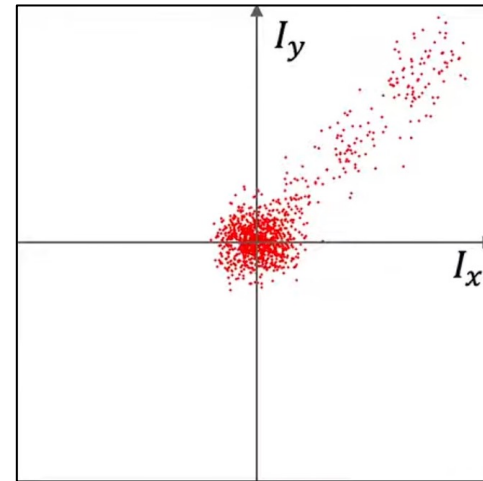
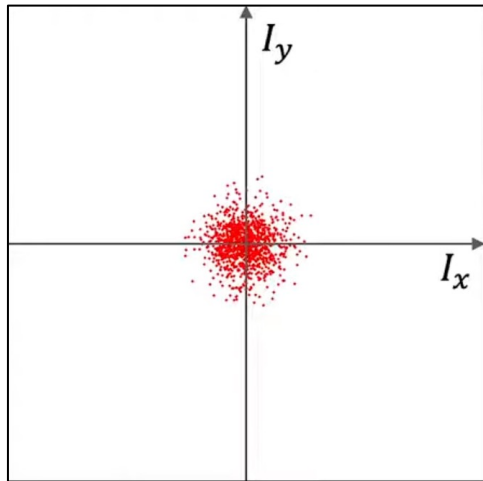
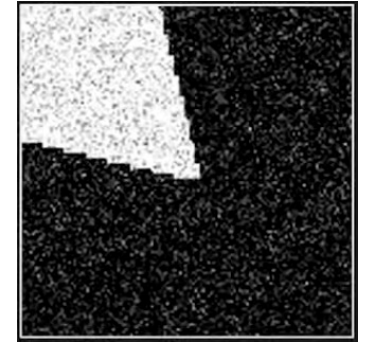
Flat



Edge

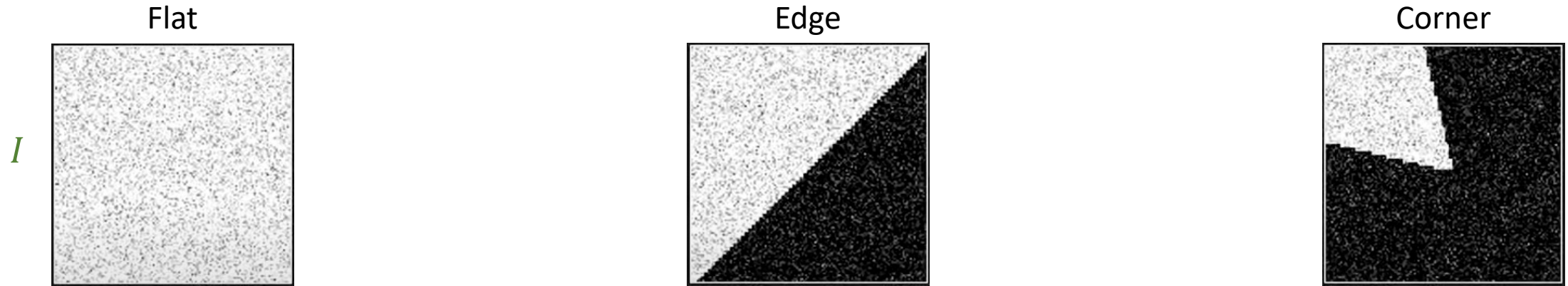


Corner



⇒ Distributions of gradients along x and y in the region is different and characterizes the type of the region.

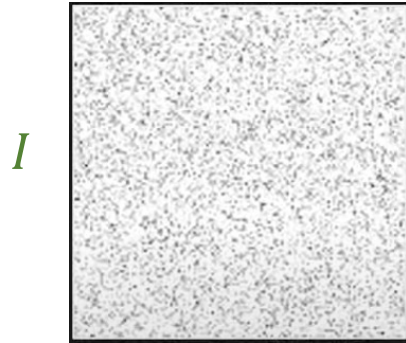
IX. Corner detection



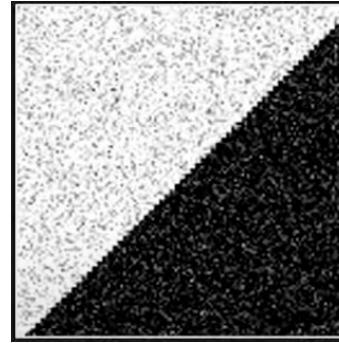
- Distributions of gradients along x and y in the region characterizes the type of the region.
- In order to automatically detect the type of the region, the key idea is to quantify this structure:
 - Describe (summarize) the distribution of gradients using a small number of parameters.
 - Use these parameters to categorize/classify the region as flat, edge or corner.

IX. Corner detection

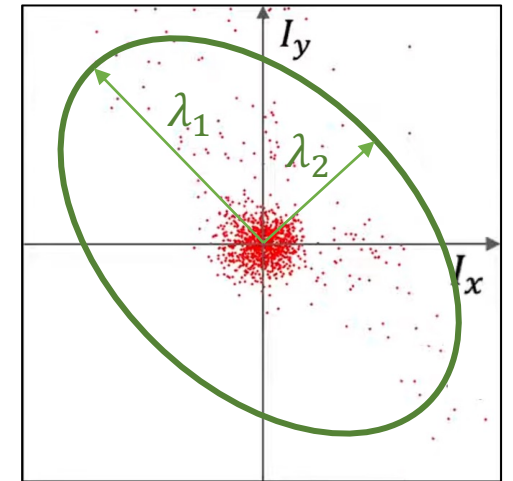
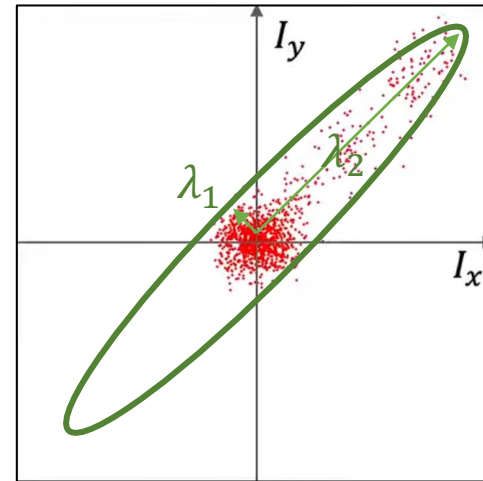
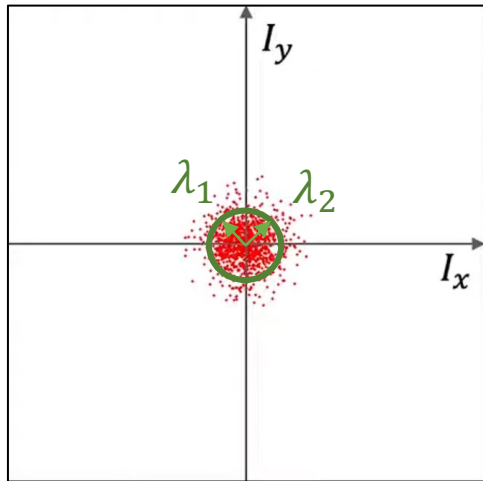
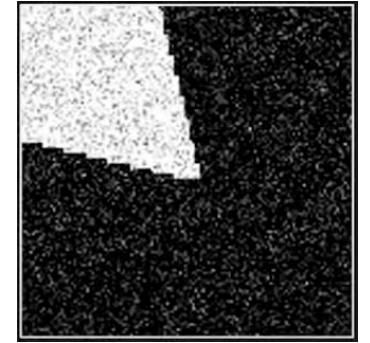
Flat



Edge

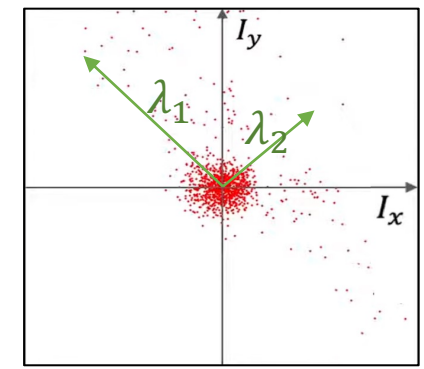
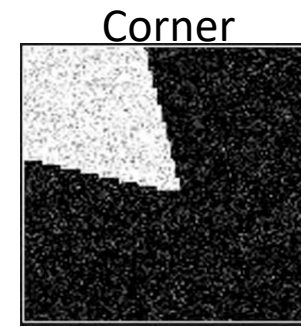


Corner



⇒ Use of λ_1 and λ_2 as parameters to categorize the region.

IX. Corner detection



- λ_1 is computed in the direction along which the inertia is maximum.
- λ_2 is computed in the direction along which the inertia is minimum.
- The problem can be expressed as a Principal Component Analysis (PCA) performed on the gradients computed in the region.
- The data to analyze is composed of a set of gradient vectors $\nabla I(x, y) = \begin{bmatrix} I_x(x, y) \\ I_y(x, y) \end{bmatrix}$, where I_x and I_y are partial derivatives (changes) in x and y directions.

IX. Corner detection: The Harris detector.

- The Harris detector relies on the following structure tensor (M):

$$M = \sum_{x,y \in \text{window}} w(x,y) \begin{bmatrix} I_x^2(x,y) & I_x(x,y)I_y(x,y) \\ I_x(x,y)I_y(x,y) & I_y^2(x,y) \end{bmatrix}$$

Where:

I_x and I_y : gradients of the image in x and y directions.

$w(x,y)$: a **weighting function**, usually a Gaussian, giving more importance to pixels near the center of the window.

- M is a **2×2 matrix** computed at **every pixel** in the image (considering that every pixel can be corner).
- But to compute M , we use a **window** (e.g., 3×3, 5×5, or Gaussian-weighted neighborhood) around the considered pixel.

IX. Corner detection: The Harris detector.

- M is *mathematically very similar* to a covariance matrix of gradient vectors in the window:

$$\text{Cov}(I_x, I_y) = \mathbb{E}[(\nabla I)(\nabla I)^T]$$

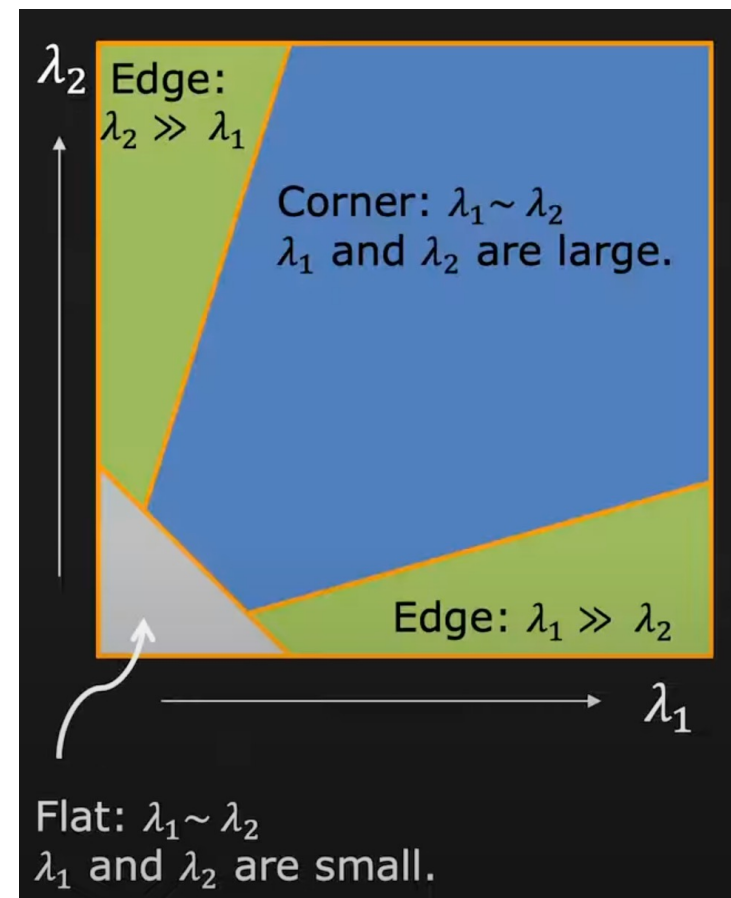
- Analysing the distribution of gradient vector around a pixel (to detect a corner) can be done by analysing the eigen values of the structure tensor M :

⇒ The eigenvalues λ_1 and λ_2 of M give the strength of the intensity change in two orthogonal directions.

IX. Corner detection: The Harris detector.

- The eigenvalues λ_1 and λ_2 of M give the strength of the intensity change in two orthogonal directions:

λ_1, λ_2	Meaning
Both small	Flat region
One large	Edge
Both large	Corner



IX. Corner detection: The Harris detector.

- The Harris detector relies on the **Corner Response Function** to compute a corner response score R :

$$R = \det(M) - k \cdot (\text{trace}(M))^2$$

where:

$\det(M) = \lambda_1 \lambda_2 \rightarrow$ product of eigenvalues of M

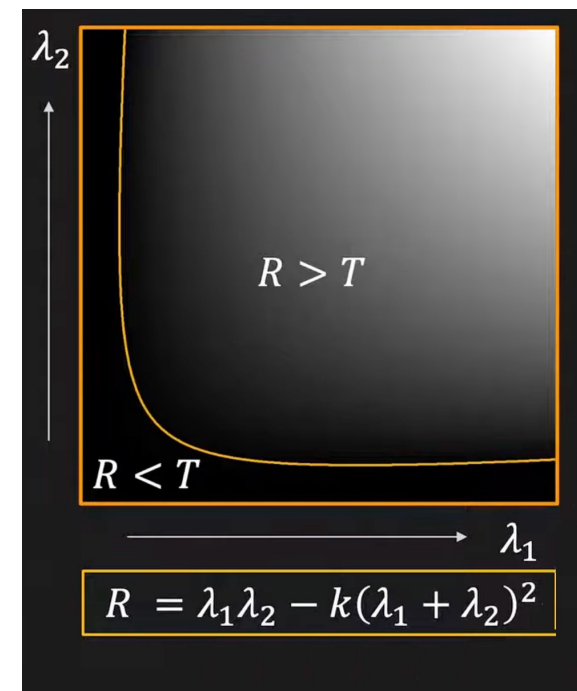
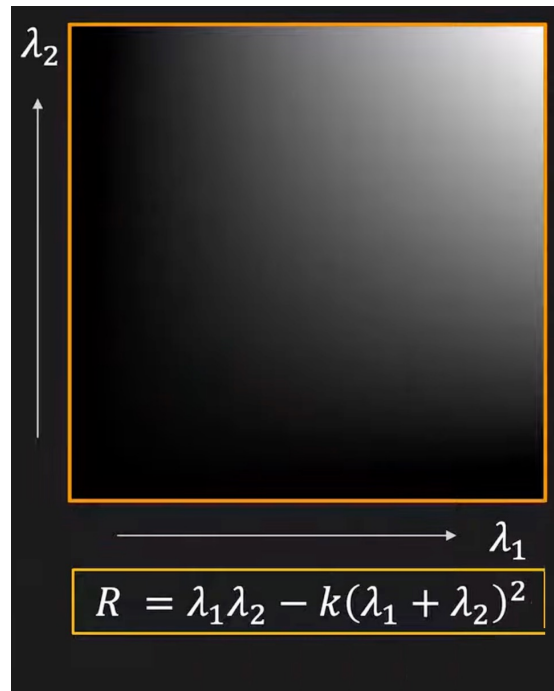
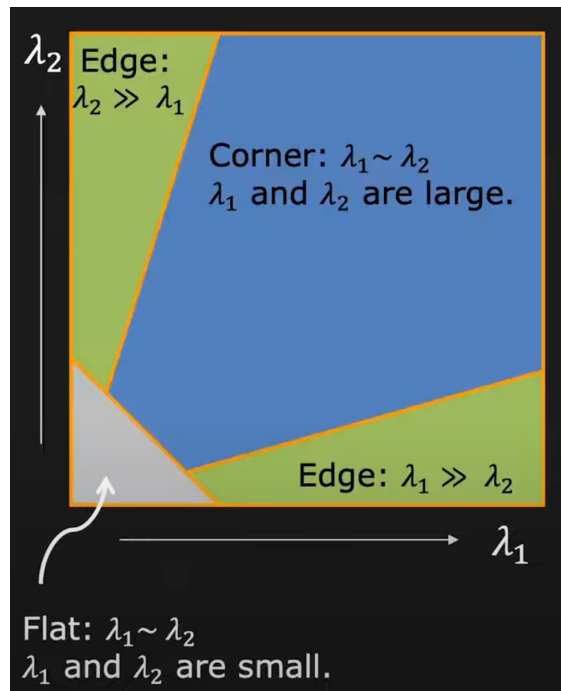
$\text{trace}(M) = \lambda_1 + \lambda_2 \rightarrow$ sum of eigenvalues

$k \in [0.04, 0.06]$ is a sensitivity constant.

- This score has been empirically designed.
- Harris is essentially performing a **PCA on local gradient vectors**, and looking for locations where **gradient energy is high in multiple directions**, which defines a corner.

IX. Corner detection: The Harris detector.

- The Harris detector relies on the **Corner Response Function**: $R = \det(M) - k \cdot (\text{trace}(M))^2$
 $\det(M) = \lambda_1 \lambda_2 \rightarrow$ product of eigenvalues of M
 $\text{trace}(M) = \lambda_1 + \lambda_2 \rightarrow$ sum of eigenvalues
 $k \in [0.04, 0.06]$ is a sensitivity constant.



IX. Corner detection: The Harris detector.



Input image



Corner response score

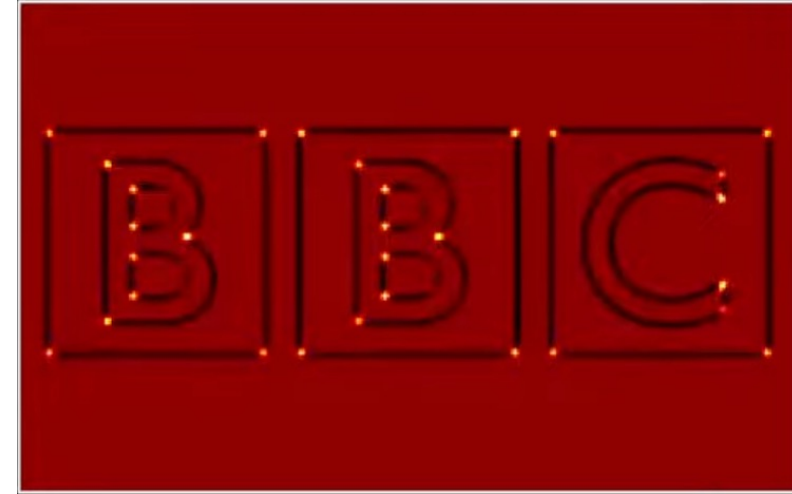


IX. Corner detection: The Harris detector.

Input image

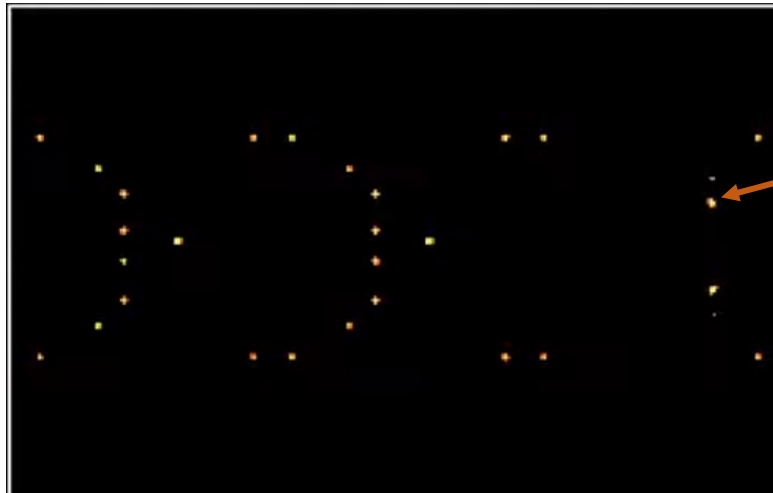


Corner response score (R)



Threshold corner response score

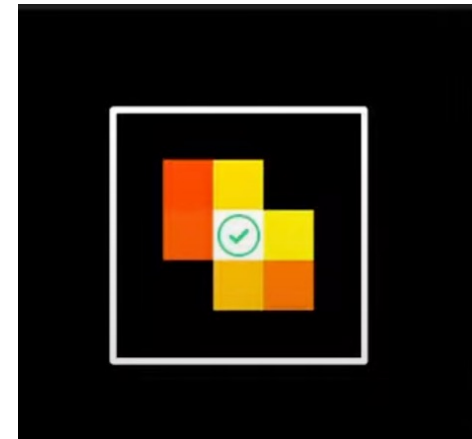
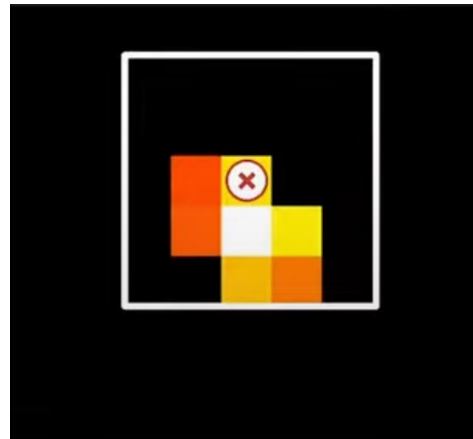
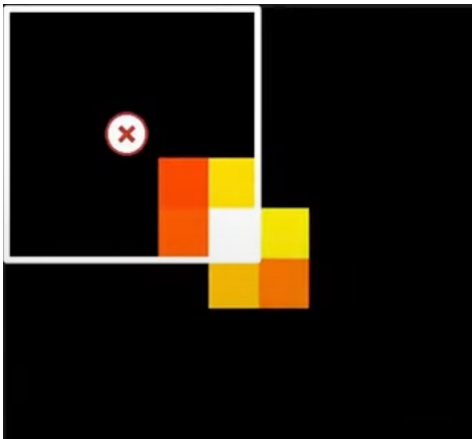
$$R > T$$



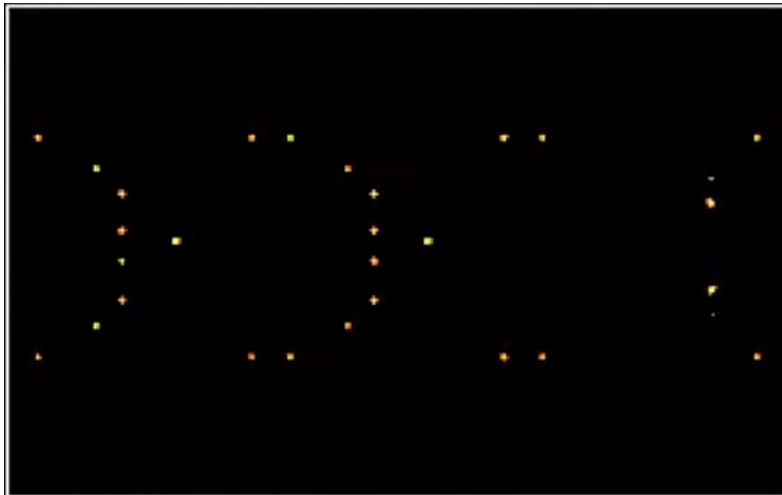
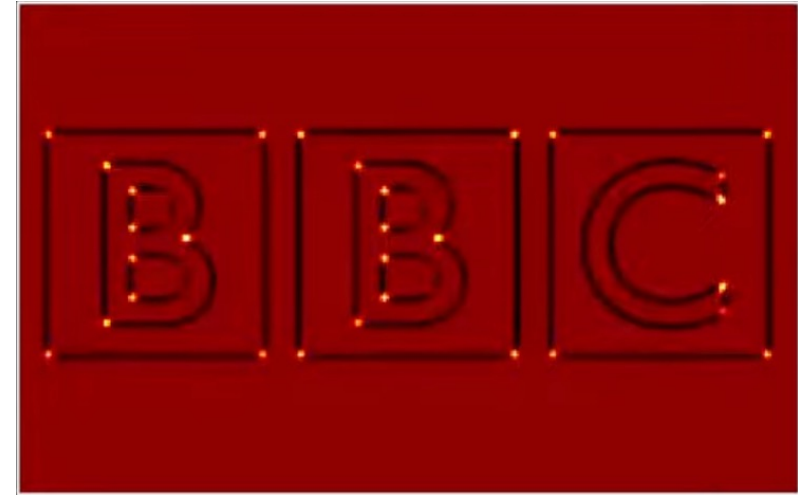
A corner is detected as a cluster of points.

IX. Corner detection: The Harris detector.

- The corner detection problem => A peak detection problem.
- One possible solution is to use the **Non-Maximal Suppression** method:
 - Slide a window of size k over the image
 - At each position, if the pixel at the center is the maximum value within the window, label it as positive (retain it). Else label it as negative (remove it)



IX. Corner detection: The Harris detector.



IX. Corner detection: The Harris detector.

The algorithm:

- Compute the image gradient: $I_x(x, y)$ and $I_y(x, y)$
- Compute the matrix M for each pixel:

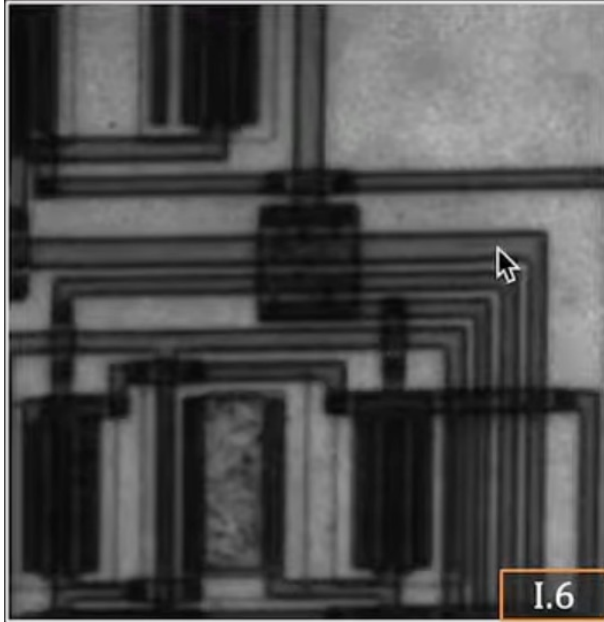
$$M = \sum_{x,y \in \text{window}} w(x, y) \begin{bmatrix} I_x^2(x, y) & I_x(x, y)I_y(x, y) \\ I_x(x, y)I_y(x, y) & I_y^2(x, y) \end{bmatrix}$$

Convolution kernel is usually Gaussian and determines the scale of the corner.

- Compute the Harris response for each pixel – the **corner response score (R)**.
- Threshold the score (R) and (optionally) perform non-maxima suppression.

IX. Corner detection: The Harris detector.

Input image



Corner response score (R)



Threshold corner response score



Detected corners



IX. Corner detection: The Harris detector.

The **Harris Corner Detector** is designed to be invariant to certain types of transformations or variations in images:

But before, let's define the invariance:

*Invariance, in the context of image processing and computer vision, refers to the ability of an algorithm or feature detector to **remain unchanged or unaffected** by specific transformations or variations applied to the input data, such as images. This means that even if the image undergoes certain modifications, the key features or detections (such as corners, edges, or objects) still remain reliable.*

- Invariance allows algorithms to **recognize features consistently**, regardless of how the image changes due to real-world factors like viewpoint, object movement, lighting, or scale.
- This is critical for tasks like object recognition, tracking, and matching, where features need to be detected and tracked across different images that may differ in transformation.

IX. Corner detection: The Harris detector.

The **Harris Corner Detector** is designed to be invariant to certain types of transformations or variations in images:

- **Translation:** The Harris detector is invariant to **translation** because the corner response R depends on the local image structure (derivatives of intensity), not on the absolute position in the image.
- **Rotation:** The Harris detector is **rotation-invariant** because it uses the **eigenvalues** of the **structure tensor** (which captures the image's local gradient information).
- **Scaling (limited):** The Harris detector is **partially invariant to scaling** but requires **multi-scale analysis** to handle large variations in size. For small scaling changes, the Harris detector performs well, as the intensity gradients change similarly under scaling.
- **Illumination Changes (in part):** The Harris detector is **somewhat invariant to illumination changes** (like lighting conditions or contrast changes). This is because the computation of gradients relies on the **relative intensity changes** rather than absolute intensity values, making it somewhat resistant to global intensity shifts. However, **significant global illumination variations** or non-linear transformations (e.g., shadows) may reduce its performance.

IX. Corner detection: The Harris detector.

Performance evaluation:

- **Repeatability (or Consistency)**

This refers to the ability of the corner detector to **reliably detect the same features** in different images or under varying conditions (e.g., transformations like scaling, rotation, or illumination changes).

It is measured by applying the corner detector to **multiple images** of the same scene (which may have been transformed or altered) and then comparing the detected corners across these images.

- **Localization Accuracy**

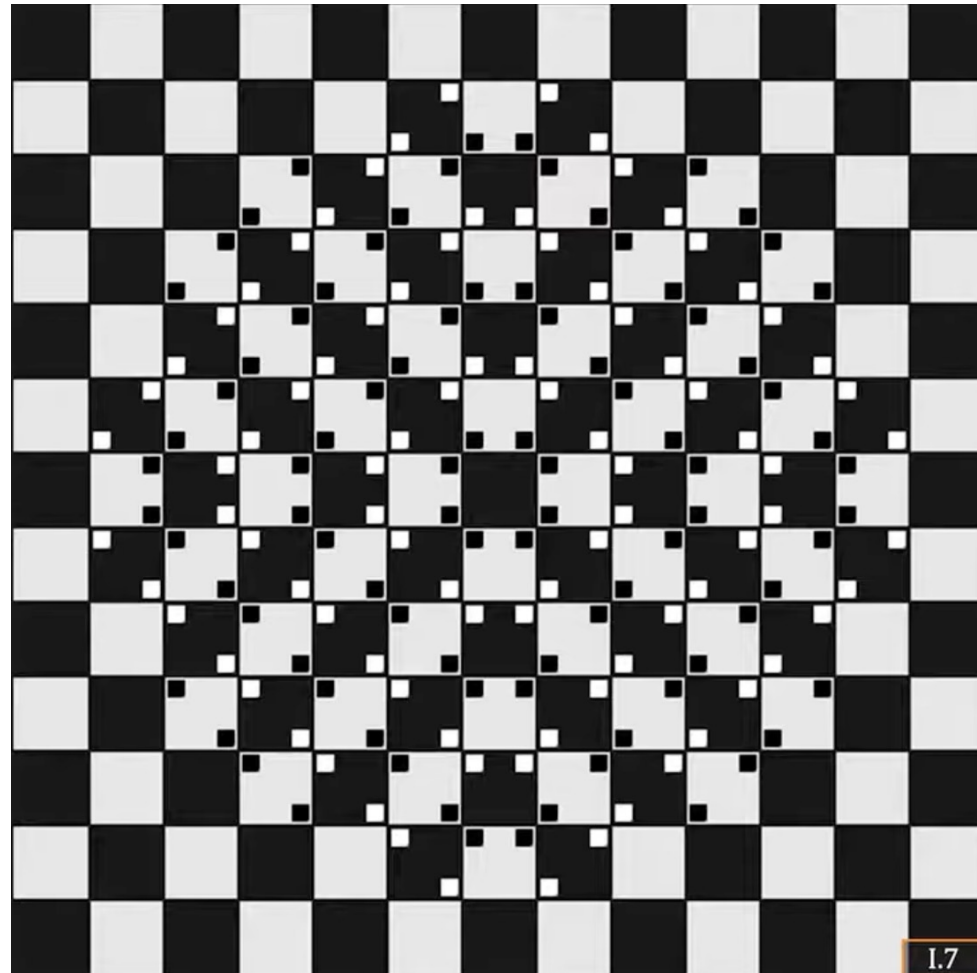
This refers to the **precision** with which the corner detector locates the exact position of the corner.

It is measured by comparing the detected corner positions to the **ground truth positions** of the corners.

- **Precision, recall...**

IX. Corner detection: The Harris detector.

- Corner Optical illusion: The bulge – Kitaoka 1998.



IX. Corner detection: The Harris detector.

⇒ Other corner detectors based on the structure tensor:

- Shi-Tomasi Corner Detector (1994)

Instead of using the Harris response R , Shi and Tomasi proposed a simpler criterion:

A point is considered a corner if the **smallest eigenvalue** $\lambda_{\min}$ of the structure tensor is large enough.

Difference from Harris: Harris combines eigenvalues into a score R ; Shi-Tomasi directly checks eigenvalues without heuristic combination.

IX. Corner detection: The Harris detector.

⇒ Other corner detectors based on the structure tensor:

- Förstner Detector (1987)

Designed for **precise feature localization** (especially in photogrammetry).

Measures two quantities: "**interest**" (how good a point is) and "**accuracy**" (how precisely the point is located).

A point is a good feature if:

Both eigenvalues are large (like Harris and Shi-Tomasi),
And their **ratio** is not too extreme (ensuring isotropy).

IX. Corner detection: The Harris detector.

<i>Detectors</i>	<i>Criterion</i>	<i>Main Idea</i>	<i>Special Feature</i>
Harris	$R = \lambda_1 \lambda_2 - k (\lambda_1 + \lambda_2)^2$	Combines eigenvalues with a heuristic	Sensitive to parameter k
Shi-Tomasi	$\min(\lambda_1, \lambda_2) > threshold$	Smallest eigenvalue must be large	Simpler, more robust for tracking
Förstner	High eigenvalues + balanced ratio	Detects and quantifies feature accuracy	Provides corner strength and precision